

The Physical Implementation of Quantum Computation

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Abstract

After a brief introduction to the principles and promise of quantum information processing, the requirements for the physical implementation of quantum computation are discussed. These five requirements, plus two relating to the communication of quantum information, are extensively explored and related to the many schemes in atomic physics, quantum optics, nuclear and electron magnetic resonance spectroscopy, superconducting electronics, and quantum-dot physics, for achieving quantum computing.

1. Introduction

The advent of quantum information processing, as an abstract concept, has given birth to a great deal of new thinking, of a very concrete form, about how to create physical computing devices that operate in the hitherto unexplored quantum mechanical regime. The efforts now underway to produce working laboratory devices that perform this profoundly new form of information processing are the subject of this book.

In this chapter I provide an overview of the common objectives of the investigations reported in the remainder of this special issue. The scope of the approaches, proposed and underway, to the implementation of quantum hardware is remarkable, emerging from specialties in atomic physics [1], in quantum optics [2], in nuclear [3] and electron [4] magnetic resonance spectroscopy, in superconducting device physics [5], in electron physics [6], and in mesoscopic and quantum dot research [7]. This amazing variety of approaches has arisen because, as we will see, the principles of quantum computing are posed using the most fundamental ideas of quantum mechanics, ones whose embodiment can be contemplated in virtually every branch of quantum physics.

The interdisciplinary spirit which has been fostered as a result is one of the most pleasant and remarkable features of this field. The excitement and freshness that has been produced bodes well for the prospect for discovery, invention, and innovation in this endeavor.

2. Why *Quantum* Information Processing?

The shortest of answers to this question would be, why not? The manipulation and transmission of information is today carried out by physical machines (computers, routers, scanners, etc.), in which the embodiment and transformations of this information can be described using the language of classical mechanics. But the final physical theory of the world is not Newtonian mechanics, and there is no reason to suppose that machines following the laws of quantum mechanics should have the same computational power as classical machines; indeed, since Newtonian mechanics emerges as a special limit of quantum mechanics, quantum machines can only have greater computational power than classical ones. The great pioneers and visionaries who pointed the way towards quantum computers, DEUTSCH [8], FEYNMAN [9], and others, were stimulated by such thoughts. Of course, by a

similar line of reasoning, it may well be asked whether machines embodying the principles of other refined descriptions of nature (perhaps general relativity or string theory) may have even more information processing capabilities; speculations exist about these more exotic possibilities, but they are beyond the scope of the present discussion.

But computing with quantum mechanics really deserves a lot more attention than worm-hole computing or quantum-gravity computing; quantum computing, while far in the future from the perspective of CMOS roadmaps and projections of chip fab advances, can certainly be seen as a real prospect from the perspective of research studies in quantum physics. It does not require science fiction to envision a quantum computer; the proposals discussed later in this issue paint a rather definite picture of what a real quantum computer will look like.

So, how much is gained by computing with quantum physics over computing with classical physics? We do not seem to be near to a final answer to this question, which is natural since even the ultimate computing power of classical machines remains unknown. But the answer as we know it today has an unexpected structure; it is not that quantum tools simply speed up all information processing tasks by a uniform amount. By a standard complexity measure (i.e., the way in which the number of computational steps required to complete a task grows with the “size” n of the task), some tasks are not sped up at all [10] by using quantum tools (e.g., obtaining the n th iterate of a function $f(f(\dots f(x)\dots))$ [11]), some are sped up moderately (locating an entry in a database of n entries [12]), and some are apparently sped up exponentially (Shor’s algorithm for factoring an n -digit number [13]).

In other types of information processing tasks, particularly those involving communication [14], both quantitative and qualitative improvements are seen [15]: for certain tasks (choosing a free day for an appointment between two parties from out of n days) there is a quadratic reduction of the amount of communicated data required, if quantum states rather than classical states are transmitted [16]. For some tasks (the “set disjointness problem”, related to allocating non-overlapping segments of a shared memory in a distributed computation) the reduction of required communication is exponential [17]. Finally, there are tasks that are doable in the quantum world that have no counterpart classically: quantum cryptography provides an absolute secrecy of communication between parties that is impossible classically [18]. And for some games, winning strategies become possible with the use of quantum resources that are not available otherwise [19, 20].

This issue, and this chapter, are primarily concerned with the “hows” of quantum computing rather than the “whys,” so we will leave behind the computer science after this extremely brief mention. There is no shortage of other places to obtain more information about these things; I recommend the recent articles by Aharonov [21] and by Cleve [22]; other general introductions [23] will give the reader pointers to the already vast specialized literature on this subject.

3. Realizing Quantum Computation

Let me proceed with the main topic: the physical realization of quantum information processing. As a guide to the remainder of the special issue, and as a means of reviewing the basic steps required to make quantum computation work, I can think of no better plan than to review a set of basic criteria that my coworkers and I have been discussing over the last few years [24] for the realization of quantum computation (and communication), and to discuss the application of these criteria to the multitude of physical implementations that are found below.

So, without further ado, here are the

Five (plus two) requirements for the implementation of quantum computation

1. A scalable physical system with well characterized qubits

For a start, a physical system containing a collection of qubits is needed. A qubit (or, more precisely, the embodiment of a qubit) is [25] simply a quantum two-level system like the two spin states of a spin 1/2 particle, like the ground and excited states of an atom, or like the vertical and horizontal polarization of a single photon. The generic notation for a qubit state denotes one state as $|0\rangle$ and the other as $|1\rangle$. The essential feature that distinguishes a qubit from a bit is that, according to the laws of quantum mechanics, the permitted states of a single qubit fills up a two-dimensional complex vector space; the general state is written $a|0\rangle + b|1\rangle$, where a and b are complex numbers, and a normalization convention $|a|^2 + |b|^2 = 1$ is normally adopted. The general state of two qubits, $a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, is a four-dimensional vector, one dimension for each distinguishable state of the two systems. These states are generically *entangled*, meaning that they cannot be written as a product of the states of two individual qubits. The general state of n qubits is specified by a 2^n -dimensional complex vector.

A qubit being “well characterized” means several different things. Its physical parameters should be accurately known, including the internal Hamiltonian of the qubit (which determines the energy eigenstates of the qubit, which are often, although not always, taken as the $|0\rangle$ and $|1\rangle$ states), the presence of and couplings to other states of the qubit, the interactions with other qubits, and the couplings to external fields that might be used to manipulate the state of the qubit. If the qubit has third, fourth, etc., levels, the computer’s control apparatus should be designed so that the probability of the system ever going into these states is small. The smallness of this and other parameters will be determined by the capabilities of quantum error correction, which will be discussed under requirement 3.

Recognizing a qubit can be trickier than one might think. For example, we might consider a pair of one-electron quantum dots that share a single electron between them as a two-qubit system. It is certainly true that we can denote the presence or absence of an electron on each dot by $|0\rangle$ and $|1\rangle$, and it is well known experimentally how to put this system into the “entangled” state $1/\sqrt{2}(|01\rangle + |10\rangle)$ in which the electron is in a superposition of being on the left dot and the right dot. But it is fallacious to consider this as a two-qubit system; while the states $|00\rangle$ and $|11\rangle$ are other allowed physical states of the dots, superselection principles forbid the creation of entangled states involving different particle numbers such as $1/\sqrt{2}(|00\rangle + |11\rangle)$.

It is therefore false to consider this as a two-qubit system, and, since there are not two qubits, it is nonsense to say that there is entanglement in this system. It is correct to say that the electron is in a superposition of different quantum states living on the two different dots. It is also perfectly correct to consider this system to be the embodiment of a *single* qubit, spanned by the states (in the misleading notation above) $|01\rangle$ (“electron on the right dot”) and $|10\rangle$ (“electron on the left dot”). Indeed, several of the viable proposals, including the ones by Schön, Averin, and Tanamoto in this special issue, use exactly this system as a qubit. However, false lines of reasoning like the one outlined here have sunk various proposals before they were properly launched (no such abortive proposals are represented in this book, but they can be found occasionally in the literature).

An amazing variety of realizations of the qubit are represented in this volume. There is a very well developed line of work that began with the proposal of CIRAC and ZOLLER [1] for an ion-trap quantum computer, in which, in its quiescent state, the computer holds the qubits in pairs of energy levels of ions held in a linear electromagnetic trap. Various pairs of energy levels (e.g., Zeeman-degenerate ground states, as are also used in the NMR approach [3] discussed by Cory) have been proposed and investigated experimentally. The many neutral-atom proposals (see chapters by KIMBLE [2], DEUTSCH [26], and BRIEGEL [27]) use similar atomic energy levels of neutral species. These atomic-physics based propos-

als use other auxiliary qubits such as the position of atoms in a trap or lattice, the presence or absence of a photon in an optical cavity, or the vibrational quanta of trapped electrons, ions or atoms (in the Platzman proposal below [6] this is the primary qubit). Many of the solid-state proposals exploit the fact that impurities or quantum dots have well characterized discrete energy level spectra; these include the spin states of quantum dots (see chapters by LOSS [7] and IMAMOGLU [2]), the spin states of donor impurities (see KANE [4]), and the orbital or charge states of quantum dots (see TANAMOTO [7]). Finally, there are a variety of interesting proposals which use the quantized states of superconducting devices, either ones involving the (Cooper-pair) charge (see SCHÖN, AVERIN), or the flux (see MOOIJ) [5].

2. The ability to initialize the state of the qubits to a simple fiducial state, such as $|000\dots\rangle$
This arises first from the straightforward computing requirement that registers should be initialized to a known value before the start of computation. There is a second reason for this initialization requirement: quantum error correction (see requirement 3 below) requires a continuous, fresh supply of qubits in a low-entropy state (like the $|0\rangle$ state). The need for a continuous supply of 0s, rather than just an initial supply, is a real headache for many proposed implementations. But since it is likely that a demonstration of a substantial degree of quantum error correction is still quite some time off, the problem of continuous initialization does not have to be solved very soon; still, experimentalists should be aware that the speed with which a qubit can be zeroed will eventually be a very important issue. If the time it takes to do this initialization is relatively long compared with gate-operation times (see requirement 4), then the quantum computer will have to be equipped with some kind of “qubit conveyor belt”, on which qubits in need of initialization are carried away from the region in which active computation is taking place, initialized while on the “belt”, then brought back to the active place after the initialization is finished. (A similar parade of qubits will be envisioned in requirement 5 for the case of low quantum-efficiency measurements [28].)

There are two main approaches to setting qubits to a standard state: the system can either be “naturally” cooled when the ground state of its Hamiltonian is the state of interest, or the standard state can be achieved by a measurement which projects the system either into the state desired or another state which can be rotated into it. These approaches are not fundamentally different from one another, since the projection procedure is a form of cooling; for instance, the laser cooling techniques used routinely now for the cooling of ion states to near their ground state in a trap [1] are closely connected to the fluorescence techniques used to measure the state of these ions. A more “natural” kind of cooling is advocated in many of the electron spin resonance based techniques (using quantum dots or impurities) [7, 4] in which the spins are placed in a strong magnetic field and allowed to align with it via interaction with their heat bath. In this kind of approach the time scale will be a problem. Since the natural thermalization times are never shorter than the decoherence time of the system, this procedure will be too slow for the needs of error correction and a “conveyor belt” scheme would be required. Cooling by projection, in which the Hamiltonian of the system and its environment are necessarily perturbed strongly, will have a time scale dependent on the details of the setup, but potentially much shorter than the natural relaxation times. One cannot say too much more at this point, as the schemes for measurement have in many cases not been fully implemented (see requirement 5). In the NMR quantum computer implementations to date (see Cory below), cooling of the initial state has been foregone altogether; it is acknowledged [3] that until some of the proposed cooling schemes are implemented (a nontrivial thing to do), NMR can never be a scalable scheme for quantum computing.

3. Long relevant decoherence times, much longer than the gate operation time
Decoherence times characterize the dynamics of a qubit (or any quantum system) in contact with its environment. The (somewhat overly) simplified definition of this time is that it is

the characteristic time for a generic qubit state $|\psi\rangle = a|0\rangle + b|1\rangle$ to be transformed into the mixture $\rho = |a|^2|0\rangle\langle 0| + |b|^2|1\rangle\langle 1|$. A more proper characterization of decoherence, in which the decay can depend on the form of the initial state, in which the state amplitudes may change as well, and in which other quantum states of the qubit can play a role (in a special form of state decay called “leakage” in quantum computing [29, 30]), is rather more technical than I want to get here; but see Refs. [31] and [32] for a good general discussion of all these. Even the simplest discussion of decoherence that I have given here should also be extended to include the possibility that the decoherence of neighboring qubits is correlated. It seems safest to assume that they will be neither completely correlated nor completely uncorrelated, and the thinking about error correction has taken this into account.

Decoherence is very important for the fundamentals of quantum physics, as it is identified as the principal mechanism for the emergence of classical behavior. For the same reason, decoherence is very dangerous for quantum computing, since if it acts for very long, the capability of the quantum computer will not be so different from that of a classical machine. The decoherence time must be long enough that the uniquely quantum features of this style of computation have a chance to come into play. How long is “long enough” is also indicated by the results of quantum error correction, which I will summarize shortly.

I have indicated that the “relevant” decoherence times should be long enough. This emphasizes that a quantum particle can have many decoherence times pertaining to different degrees of freedom of that particle. But many of these can be irrelevant to the functioning of this particle as a qubit. For example, the rapid decoherence of an electron’s position state in a solid state environment does not preclude its having a very long spin coherence time, and it can be arranged that this is the only time relevant for quantum computation. Which time is relevant is determined by the choice of the qubit basis states $|0\rangle$ and $|1\rangle$; for example, if these two states correspond to different spin states but identical orbital states, then orbital decoherence will be irrelevant.

One might worry that the decoherence time necessary to do a successful quantum computation will scale with the duration of the computation. This would place incredibly stringent requirements on the physical system implementing the computation. Fortunately, in one of the great discoveries of quantum information theory (in 1995–1996), it was found that error correction of quantum states is possible [33] and that this correction procedure can be successfully applied in quantum computation [34], putting much more reasonable (although still daunting) requirements on the needed decoherence times.

In brief, quantum error correction starts with coding; as in binary error correction codes, in which only a subset of all boolean strings are “legal” states, quantum error correction codes consist of legal states confined to a subspace of the vector space of a collection of qubits. Departure from this subspace is caused by decoherence. Codes can be chosen such that, with a suitable sequence of quantum computations and measurements of some ancillary qubits, the error caused by decoherence can be detected and corrected. As noted above, these ancillary qubits have to be continuously refreshed for use. I will not go much farther into the subject here, see [31] for more. It is known that quantum error correction can be made fully fault tolerant, meaning that error correction operations can be successfully intermingled with quantum computation operations, that errors occurring during the act of error correction, if they occur at a sufficiently small rate, do no harm, and that the act of quantum computation does not itself cause an unreasonable proliferation of errors.

These detailed analyses have indicated the magnitude of decoherence time scales that are acceptable for fault-tolerant quantum computation. The result is that, if the decoherence time is 10^4 – 10^5 times the “clock time” of the quantum computer, that is, the time for the execution of an individual quantum gate (see requirement 4), then error correction can be successful. This is, to tell the truth, a rather stringent condition, quantum systems frequently do not have such long decoherence times. But sometimes they do, and our search for a successful physical implementation must turn towards these. At least this result says that

the required decoherence rate does not become ever smaller as the size and duration of the quantum computation grows. So, once the desired threshold is attainable, decoherence will not be an obstacle to scalable quantum computation.

Having said this, it must be admitted that it will be some time before it is even possible to subject quantum error correction to a reasonable test. Nearly all parts of requirements 1–5 must be in place before such a test is possible. And even the most limited application of quantum error correction has quite a large overhead: roughly 10 ancillary qubits must be added for each individual qubit of the computation. Fortunately, this overhead ratio grows only logarithmically as the size of the quantum computation is increased.

In the short run, it is at least possible to design and perform experiments which measure the decoherence times and other relevant properties (such as the correlation of decoherence of neighboring qubits) of candidate implementations of qubits. With such initial test experiments, caution must be exercised in interpreting the results, because decoherence is a very system-specific phenomenon, depending on the details of all the qubits' couplings to various environmental degrees of freedom. For example, the decoherence time of the spin of an impurity in the bulk of a perfect semiconductor may not be the same as its decoherence time when it is near the surface of the solid, in the immediate neighborhood of device structures designed to manipulate its quantum state. Test experiments should probe decoherence in as realistic a structure as is possible.

4. A “universal” set of quantum gates

This requirement is of course at the heart of quantum computing. A quantum algorithm is typically specified [8] as a sequence of unitary transformations $U_1, U_2, U_3, \dots$, each acting on a small number of qubits, typically no more than three. The most straightforward transcription of this into a physical specification is to identify Hamiltonians which generate these unitary transformations, viz., $U_1 = e^{iH_1 t/\hbar}$, $U_2 = e^{iH_2 t/\hbar}$, $U_3 = e^{iH_3 t/\hbar}$, etc.; then, the physical apparatus should be designed so that H_1 can be turned on from time 0 to time t , then turned off and H_2 turned on from time t to time $2t$, etc.

Would that life were so simple! In reality what can be done is much less, but much less can be sufficient. Understanding exactly how much less is still enough, is the main complication of this requirement. In all the physical implementations discussed in this volume, only particular sorts of Hamiltonians can be turned on and off; in most cases, for example, only two-body (two-qubit) interactions are considered. This immediately poses a problem for a quantum computation specified with three-qubit unitary transformations; fortunately, of course, these can always be re-expressed in terms of sequences of one- and two-body interactions [35], and the two-body interactions can be of just one type [36], the “quantum XOR” or “cNOT”. There are some implementations in which multi-qubit gates can be implemented directly [37].

However, this still leaves a lot of work to do. In some systems, notably in NMR (see Cory), there are two-body interactions present which cannot be turned *off*, as well as others which are switchable. This would in general be fatal for quantum computation, but the particular form of the fixed interactions permit their effects to be annulled by particular “refocusing” sequences of the controllable interactions, and it has recently been discovered [38] that these refocusing sequences can be designed and implemented efficiently.

For many other systems, the two-body Hamiltonian needed to generate directly the cNOT unitary transformation is not available. For example, in the quantum-dot proposal described by Loss below [7], the only two-body interaction which should be easily achievable is the exchange interaction between neighboring spins, $H \propto \vec{S}_i \cdot \vec{S}_{i+1}$; in the Imamoglu chapter [2], the attainable interaction is of the XY type, i.e., $H \propto S_{ix}S_{jx} + S_{iy}S_{jy}$. An important observation is that with the appropriate sequence of exchange or XY interactions, in conjunction with particular one-body interactions (which are assumed to be more easily doable), the cNOT transformation can be synthesized [39]. It is incumbent on each implementation pro-

posal to exhibit such a sequence for producing the cNOT using the interactions that are naturally realizable.

Often there is also some sophisticated thinking required about the time profile of the two-qubit interaction. The naive description above uses a “square pulse” time profile, but often this is completely inappropriate; for instance, if the Hamiltonian can also couple the qubit to other, higher-lying levels of the quantum system, often the only way to get the desired transformation is to turn on and off the interaction smoothly and slowly enough that an adiabatic approximation is accurate [29, 30] (in a solid-state context, see also [40]). The actual duration of the pulse will have to be sufficiently long that any such adiabatic requirement is satisfied; then typically only the time integral $\int dt H(t)$ is relevant for the quantum gate action. The overall time scale of the interaction pulse is also controlled by the attainable maximum size of the matrix elements of $H(t)$, which will be determined by various fundamental considerations, like the requirement that the system remains in the regime of validity of a linear approximation, and practical considerations, like the laser power that can be concentrated on a particular ion. Given these various constraints, the “clock time” of the quantum computer will be determined by the time interval needed such that two consecutive pulses have negligible overlap.

Another consideration, which does not seem to present a problem with any current implementation schemes, but which may be an issue in the future, is the classicality of the control apparatus. We say that the interaction Hamiltonian $H(t)$ has a time profile which is controlled externally by some “classical” means, that is, by the intensity of a laser beam, the value of a gate voltage, or the current level in a wire. But each of these control devices is made up themselves of quantum mechanical parts. When we require that these behave classically, it means that their action should proceed without any entanglement developing between these control devices and the quantum computer. Estimates indicate that this entanglement can indeed be negligible, but this effect needs to be assessed for each individual case.

In many cases it is impossible to turn on the desired interaction between a pair of qubits; for instance, in the ion-trap scheme, no direct interaction is available between the ion-level qubits [1]. In this and in other cases, a special quantum subsystem (sometimes referred to as a “bus qubit”) is used which can interact with each of the qubits in turn and mediate the desired interaction: for the ion trap, this is envisioned to be the vibrational state of the ion chain in the trap; in other cases it is a cavity photon whose wavefunction overlaps all the qubits. Unfortunately, this auxiliary quantum system introduces new channels for the environment to couple to the system and cause decoherence, and indeed the decoherence occurring during gate operation is of concern in the ion-trap and cavity-quantum electrodynamics schemes.

Some points about requirement 4 are important to note in relation to the implementation of error correction. Successful error correction requires fully parallel operation, meaning that gate operations involving a finite fraction of all the qubits must be doable simultaneously. This can present a problem with some of the proposals in which the single “bus qubit” is needed to mediate each interaction. On the other hand, the constraint that interactions are only among nearest neighbors in a lattice, as in many of the solid-state proposals, does allow for sufficient parallelism [41].

Quantum gates cannot be implemented perfectly; we must expect both systematic and random errors in the implementation of the necessary Hamiltonians. Both types of errors can be viewed as another source of decoherence and thus error correction techniques are effective for producing reliable computations from unreliable gates, if the unreliability is small enough. The tolerable unreliability due to random errors is in the same vicinity as the decoherence threshold, that is, the magnitude of random errors should be 10^{-4} – 10^{-5} per gate operation or so. It might be hoped that systematic errors could be virtually eliminated by careful calibration; but this will surely not always be the case. It seems harder to give a

good rule for how much systematic error is tolerable, the conservative estimates give a very, very small number (the square of the above) [31], but on the other hand there seems to be some evidence that certain important quantum computations (e.g., the quantum Fourier transform) can tolerate a very high level of systematic error (over- or under-rotation). Some types of very large errors may be tolerable if their presence can be detected and accounted for on the fly (we are thinking, for example, about charge switching in semiconductors or superconductors).

Error correction requires that gate operations be done on coded qubits, and one might worry that such operations would require a new repertoire of elementary gate operations for the base-level qubits which make up the code. For the most important error correction techniques, using the so called “stabilizer” codes, this is not the case. The base-level toolkit is exactly the same as for the unencoded case: one-bit gates and cNOTs, or any gate repertoire that can produce these, are adequate. Sometimes the use of coding can actually *reduce* the gate repertoire required: in the work on decoherence free subspaces and subsystems, codes are introduced using blocks of three and four qubits for which two-qubit exchange interactions alone are enough to implement general quantum computation [42, 43]. This simplification could be very useful in the quantum-dot [7] or semiconductor-impurity [4] implementations.

5. A qubit-specific measurement capability

Finally, the result of a computation must be read out, and this requires the ability to measure specific qubits. In an ideal measurement, if a qubit’s density matrix is $\rho = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1| + \alpha|0\rangle\langle 1| + \alpha^*|1\rangle\langle 0|$, the measurement should give outcome “0” with probability p and “1” with probability $1-p$ independent of α and of any other parameters of the system, including the state of nearby qubits, and without changing the state of the rest of the quantum computer. If the measurement is “non-demolition”, that is, if in addition to reporting outcome “0” the measurement leaves the qubit in state $|0\rangle\langle 0|$, then it can also be used for the state preparation of requirement 2; but requirement 2 can be fulfilled in other ways.

Such an ideal measurement as I have described is said to have 100% quantum efficiency; real measurements always have less. While the fidelity of a quantum measurement is not captured by a single number, the single quantum-efficiency parameter is often a very useful way to summarize it, just as the decoherence time is a useful if incomplete summary of the damage caused to a quantum state by the environment.

While quantum efficiency of 100% is desirable, much less is needed for quantum computation; there is, in fact, a tradeoff possible between quantum efficiency and other resources which results in reliable computation. As a simple example, if the quantum efficiency is 90%, then, in the absence of any other imperfections, a computation with a single-bit output (a so-called “decision problem”, common in computer science) will have 90% reliability. If 97% reliability is needed, this can just be achieved by rerunning the calculation three times. Much better, actually, is to “copy” the single output qubit to three, by applying two cNOT gates involving the output qubit and two other qubits set to $|0\rangle$, and measuring those three. (Of course, qubits cannot be “copied”, but their value in a particular basis can.) In general, if quantum efficiency q is available, then copying to somewhat more than $1/q$ qubits and measuring all of these will result in a reliable outcome. So, a quantum efficiency of 1% would be usable for quantum computation, at the expense of hundreds of copies/remasures of the same output qubit. (This assumes that the measurement does not otherwise disturb the quantum computer. If it does, the possibilities are much more limited.)

Even quantum efficiencies much, much lower than 1% can be and are used for successful quantum computation: this is the “bulk” model of NMR (see Cory and [3]), where macroscopic numbers of copies of the same quantum computer (different molecules in solu-

tion) run simultaneously, with the final measurement done as an ensemble average over the whole sample. These kinds of weak measurements, in which each individual qubit is hardly disturbed, are quite common and well understood in condensed-matter physics.

If a measurement can be completed quickly, on the timescale of 10^{-4} of the decoherence time, say, then its repeated application during the course of quantum computation is valuable for simplifying the process of quantum error correction. On the other hand, if this fast measurement capability is not available, quantum error correction is still possible, but it then requires a greater number of quantum gates to implement.

Other tradeoffs between the complexity and reliability of quantum measurement vs. those of quantum computation have recently been explored. It has been shown that if qubits can be initialized into pairs of maximally entangled states, and two-qubit measurements in the so-called Bell basis ($\Psi^\pm = |01\rangle \pm |10\rangle$, $\Phi^\pm = |00\rangle \pm |11\rangle$) are possible, then no two-qubit quantum gates are needed, one-bit gates alone suffice [44]. Now, often this tradeoff will not be useful, as in many schemes a Bell measurement would require two-bit quantum gates.

But the overall message, seen in many of our requirements, is that more and more, the theoretical study of quantum computation has offered a great variety of tradeoffs for the potential implementations: if X is very hard, it can be substituted with more of Y. Of course, in many cases both X and Y are beyond the present experimental state of the art; but a thorough knowledge of these tradeoffs should be very useful for devising a rational plan for the pursuit of future experiments.

4. Desiderata for Quantum Communication

For computation alone, the five requirements above suffice. But the advantages of quantum information processing are not manifest solely, or perhaps even principally, for straightforward computation only. There are many kinds of information-processing tasks, reviewed briefly at the beginning, that involve more than just computation, and for which quantum tools provide a unique advantage.

The tasks we have in mind here all involve not only computation but also communication. The list of these tasks that have been considered in the light of quantum capabilities, and for which some advantage has been found in using quantum tools, is fairly long and diverse: it includes secret key distribution, multiparty function evaluation as in appointment scheduling, secret sharing, and game playing [14].

When we say communication we mean quantum communication: the transmission of intact qubits from place to place. This obviously adds more features that the physical apparatus must have to carry out this information processing. We formalize these by adding two more items to the list of requirements:

6. The ability to interconvert stationary and flying qubits

7. The ability to faithfully transmit flying qubits between specified locations

These two requirements are obviously closely related, but it is worthwhile to consider them separately, because some tasks need one but not the other. For instance, quantum cryptography [18] involves only requirement 7; it is sufficient to create and detect flying qubits directly.

I have used the jargon “flying qubits” [2], which has become current in the discussions of quantum communication. Using this term emphasizes that the optimal embodiment of qubits that are readily transmitted from place to place is likely to be very different from the optimal qubits for reliable local computation. Indeed, almost all proposals assume that photon states, with the qubit encoded either in the polarization or in the spatial wavefunction of the photon, will be the flying qubit of choice, and indeed, the well developed tech-

nology of light transmission through optical fibers provides a very promising system for the transmission of qubits. I would note, though, that my colleagues and I have raised the possibility that electrons traveling through solids could provide another realization of the flying qubit [14, 45].

Only a few completely developed proposals exist which incorporate requirements 6 and 7. Of course, there are a number of quite detailed studies of 7, in the sense that experiments on quantum cryptography have been very concerned with the preservation of the photon quantum state during transmission through optical fibers or through the atmosphere. However, these studies are rather disconnected from the other concerns of quantum computing. Requirement 6 is the really hard one; to date the only theoretical proposal sufficiently concrete that experiments addressing it have been planned is the scheme produced by Kimble and coworkers [46] for unloading a cavity photon into a traveling mode via atomic spectroscopy, and loading it by the time-reversed process. Other promising concepts, like the launching of electrons from quantum dots into quantum wires such that the spin coherence of the electrons is preserved, need to be worked out more fully.

5. Summary

So, what is the “winning” technology going to be? I don’t think that any living mortal has an answer to this question, and at this point it may be counterproductive even to ask it. Even though we have lived with quantum mechanics for a century, our study of quantum effects in complex artificial systems like those we have in mind for quantum computing is in its infancy. No one can see how or whether all the requirements above can be fulfilled, or whether there are new tradeoffs, not envisioned in our present theoretical discussions but suggested by further experiments, that might take our investigations in an entirely new path.

Indeed, the above discussion, and the other chapters of this special issue, really do not cover all the foreseeable approaches. I will mention two of which I am aware: first, another computational paradigm, that of the cellular automaton, is potentially available for exploitation. This is distinguished from the above “general purpose” approach in that it assumes that every bit pattern throughout the computer will be subjected to the same evolution rule. It is known that general-purpose computation is performable, although with considerable overhead, by a cellular automaton. This is true as well for the quantum version of the cellular automaton, as LLOYD [47] indicated in his original work. New theoretical work by BENJAMIN [48] shows very explicitly how relatively simple local rules would permit the implementation of some quantum computations. This could point us perhaps towards some sort of polymer with a string of qubits on its backbone that can be addressed globally in a spectroscopic fashion. Experiments are not oriented towards this at the moment, but the tradeoffs are very different, and I don’t believe it should be excluded in the future.

Second, even more speculative, but very elegant, is the proposal of KITAEV [49] to use quantum systems with particular kinds of topological excitations, for example nonabelian anyons, for quantum computing. It is hard to see at the moment how to turn this exciting proposal into an experimental program, as no known physical system is agreed to have the appropriate topological excitations. But further research in, for example, the quantum Hall effect might reveal such a system; more likely, perhaps, is that further understanding of this approach, and that of Freedman and his colleagues [50], will shed more light on doing quantum computing using the “standard” approach being considered in this book.

I am convinced of one thing: the ideas of quantum information theory will continue to exert a decisive influence on the further investigation of the fundamental quantum properties of complex quantum systems, and will stimulate many creative and exciting developments for many years to come.

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